

RESERVE NETWORK DESIGN COMBINING SPATIAL GRAPH THEORY AND SPECIES' SPATIAL REQUIREMENTS

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Management decisions regarding conservation reserve design are dependent on our ability to characterize landscape spatial heterogeneity and its effects on species biodiversity and persistence. Understanding species' spatial habitat requirements in fragmented forested landscapes can increase our ability to maintain species biodiversity at the landscape scale. We combine spatial graph theory and species' spatial requirements to identify networks of patches that maximize habitat connectivity in multi-use forested landscapes with a fixed conservation area. A case study is presented to illustrate the utility of spatial graph and geomatics approaches to reserve design using boreal forest data from Quebec. This research demonstrates that spatial graph based connectivity analysis can help support well-informed landscape level management decisions.

Les décisions administratives concernant la conception des réserves de conservation dépendent de notre capacité à caractériser l'hétérogénéité spatiale du paysage et ses effets sur la biodiversité et la persistance des espèces. La compréhension des besoins en terme d'habitat spatial des espèces dans les paysages boisés fragmentés peut augmenter notre capacité à maintenir la biodiversité des espèces à l'échelle du paysage. Nous combinons la théorie des graphes spatiaux et les besoins spatiaux des espèces pour déterminer des réseaux de parcelles qui optimisent la connectivité des habitats dans des paysages boisés polyvalents avec une zone de conservation fixe. Une étude de cas est présentée pour illustrer l'utilité des approches des graphes spatiaux et de la géomatique pour la conception des réserves à l'aide de données sur la forêt boréale du Québec. Cette recherche démontre que l'analyse de la connectivité basée sur les graphes spatiaux peut aider à soutenir des décisions éclairées sur la gestion du paysage.

Introduction

Habitat loss and fragmentation caused by land cover changes is the single largest threat to global biodiversity [Sala *et al.* 2000]. It is widely recognized that the maintenance of biodiversity is essential to sustainable long-term functioning of natural ecosystem processes [UNEP 1992; Loreau *et al.* 2001]. A central goal of conservation biology is to safeguard against further losses of existing biodiversity through the designation of land parcels to form networks of ecologically representative nature reserves [Meir *et al.* 2004]. Given economic and socio-political constraints, it is not feasible to conserve all areas of conservation interest. Conservation planning therefore requires spatial decision tools to prioritize areas based on their biodiversity value under the constraint of available resources [Kelley *et al.* 2002]. Many different methods have been proposed to select areas to include in a reserve network, often with the goal of representing the greatest

amount of biodiversity within a minimum area or number of sites with the least economic cost. Examples of recent approaches include heuristics and iterative algorithms based on rule sets [Pressey and Nicholls 1989; Kershaw *et al.* 1994], integer linear programming and optimality algorithms [Willis *et al.* 1996; Rodrigues *et al.* 2000], and simulated annealing [Possingham *et al.* 2000]. However, new methods are needed that deal explicitly with the spatial arrangement of these reserves to maintain a sustainable network through time.

Empirical and modelling studies have shown that species persistence in fragmented landscapes depends on the ability of organisms to move among resource patches [Fahrig and Merriam 1985; Lefkovitch and Fahrig 1985; Burel 1989; Henein and Merriam 1990; Henein *et al.* 1998]. The degree to which a landscape facilitates or impedes movement of organisms among resource patches is referred to



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as landscape connectivity [Taylor *et al.* 1993; Tischendorf and Fahrig 2000]. This property of the landscape is dependent upon the interaction between landscape spatial structure (structural connectivity) and the movement capabilities of the organism (functional connectivity) [Tischendorf and Fahrig 2000; Goodwin and Fahrig 2002a]. Hence, to effectively maintain biodiversity through time, reserve networks need to be designed with explicit consideration of habitat composition and landscape connectivity [Briers 2002; Cabeza 2003].

For these reasons, reserve selection procedures have increasingly recognized the importance of quantifying connectivity when considering the spatial configuration of reserve areas. However, geomatics tools to implement and evaluate landscape spatial properties, such as connectivity, have been lacking. The research reported here aims to provide tools that can help managers understand animal habitat requirements and movement abilities affecting species persistence in fragmented landscapes. We present a geomatics-based method to assist in the selection of spatial reserve networks. The two goals of this method are to (a) identify reserve areas that satisfy spatial biodiversity representation goals, and (b) ensure biodiversity persistence, by selecting from among identified suitable habitat patches, the set that maintains the greatest degree of connectivity at multiple scales. To achieve our goals, we: (1) develop a species spatial filter approach where species' spatial requirements (habitat size and spatial arrangement) are used to parametrize operational spatial algorithms that produce a map of potential patches for protection; and (2) determine which subset of patches optimizes the spatial distances among patches given species'

movement abilities and the targeted amount of protected area to maintain. To illustrate this new spatial method, we apply it to a multi-use management area in the mixed boreal forest of Quebec using three species. Spatial data manipulation and overlays are performed using the GIS application IDRISI [Clark University 2001] and, unless otherwise stated, all spatial models were built using the Spatially Explicit Landscape Event Simulator (SELES) modelling tool [Fall and Fall 2001].

Boreal Forest: A Case Study

Sustainable forest management in the boreal forest requires that both timber yield and biodiversity be maintained through time. With this goal in mind, it has been recognized that a proportion of any given management unit should be protected for wildlife conservation [Hunter and Calhoun 1996]. For illustration purposes, we chose a conservation target of between 10% and 12% of the study area [Noss 1996; Soulé and Sanjayan 1998; Messier and Kneeshaw 1999]. Our study area is a forest management unit in the boreal region of Vermillion (Haute Mauricie) in south central Quebec (approximately 392 000 ha; Figure 1). Data layers for this region were derived from the third decadal forest inventory (1990's) carried out by SIFORT (*Système d'Information FORestière par Tesselle*) in collaboration with the Quebec Ministry of Natural Resources and Abitibi Consolidated. These data were derived from aerial photography and field sampling and were represented in a raster format at a resolution of 0.25 ha (50 m x 50 m) for a total of 1 568 000 pixels in the study area.

The Vermillion study area contains two separate ecotypes: (1) the northern portion of the study area is characteristic of the coniferous boreal forest containing black spruce, balsam fir, and yellow birch; (2) the southern portion is a mixed-wood forest that can be described as a transition zone between the boreal and temperate Nordic zones. Characteristic species of this region include white spruce, balsam fir, and white birch [Government of Quebec 2003]. Data layers of forest composition and stand age were used to identify potential species habitat.

Species Selection

To illustrate the reserve selection procedure, we selected three species with different spatial scales of response to forest fragmentation (home range size) and different but overlapping habitat requirements (forest composition, age) [Farley 2002; Table 1].

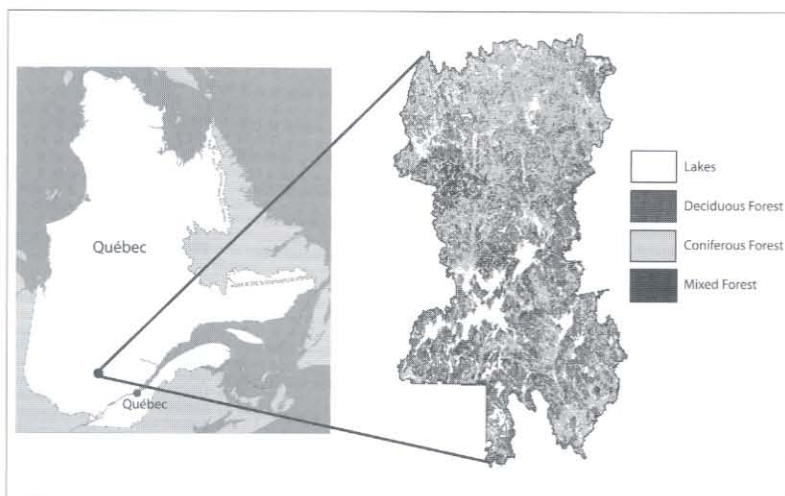


Figure 1. Location of the Vermillion study area in the province of Quebec, Canada. The map of the study area (390 000 ha) shows the spatial arrangement of habitat types at a resolution of 0.25 ha/cell.

Table 1: Species' habitat requirements.

Species	Stand Type	Stand Age	Home Range Size (HRSIZEMax)	Dispersal Distance	Source
Brown Creeper (<i>Certhia americana</i>)	Coniferous or Mixed	Mature to Old ≥60 years.	4 ha / couple	226 m	[Farley 2002]
Pileated Woodpecker (<i>Dryocopus pileatus</i>)	Deciduous or Mixed	Mature to Old ≥60 years.	200 ha / couple	1572 m	[Farley 2002] [Lafleur and Blanchette 1993]
American Marten (<i>Martes americana</i>)	Coniferous or Mixed	> 30 years.	1000 ha / individual	3500 m	[Potvin 1998] [Farley 2002]

This is a coarse filter approach to reserve site selection that assumes selecting reserves based on the spatial habitat requirements of a few species will facilitate the conservation of other components of biodiversity as well [Lambeck 1997; Margules and Pressey 2000]. The selected species in order of increasing home range size requirements were: the Brown creeper (*Certhia americana*) (2 to 4 ha), the Pileated woodpecker (*Dryocopus pileatus*) (100 to 200 ha), and the American marten (*Martes americana*) (500 to 1000 ha).

Objective 1: Identify Potential Home Range Areas using the Species Spatial Filter

Our approach to identifying species' home ranges makes several assumptions about how organisms interact with the landscape to identify and prioritize areas for membership within a reserve network. The first is that forest age and stand type are sufficient to identify areas of suitable wildlife habitat (i.e., we focus on current suitability, and not capability, which might require restoration). Also, we assume that wildlife species have specific spatial home range requirements that can be used to identify priority areas for conservation. Furthermore, home range quality can be characterized by the density of suitable habitat within that

home range area. Finally, we assume that an animal may occupy a home range of less than maximum size, provided there is a sufficient density of habitat within that home range. In this example we define "sufficient" as being an area of habitat equal to or greater than 50% of the maximum home range size [Hoover and Wills 1984; Schultz and Joyce 1992], but if available, detailed species information could be used to define minimum home range requirements and maximum home range size. For example, if a species has a maximum home range size of 100 ha, any area up to 100 ha that contains at least 50 ha of habitat is considered a viable home range. This value was adopted from recommendations on marten management based on observed habitat use in fragmented boreal landscapes [Potvin *et al.* 2000]. We apply this home range density criterion to all species selected.

Species Spatial Filter Algorithm

The aim of the species spatial filter algorithm is to identify forest patches that satisfy criteria for species' habitat types and home range size requirements to be considered as potential reserve patches (Figure 2). Such forest patches were identified in three steps: (1) create a suitable habitat layer (map) based on species forest age and stand type preferences; (2) create a home range density layer by measuring suitable habitat in a variable-size window up to the maximum size of a species home range; and (3) overlap species' home range density

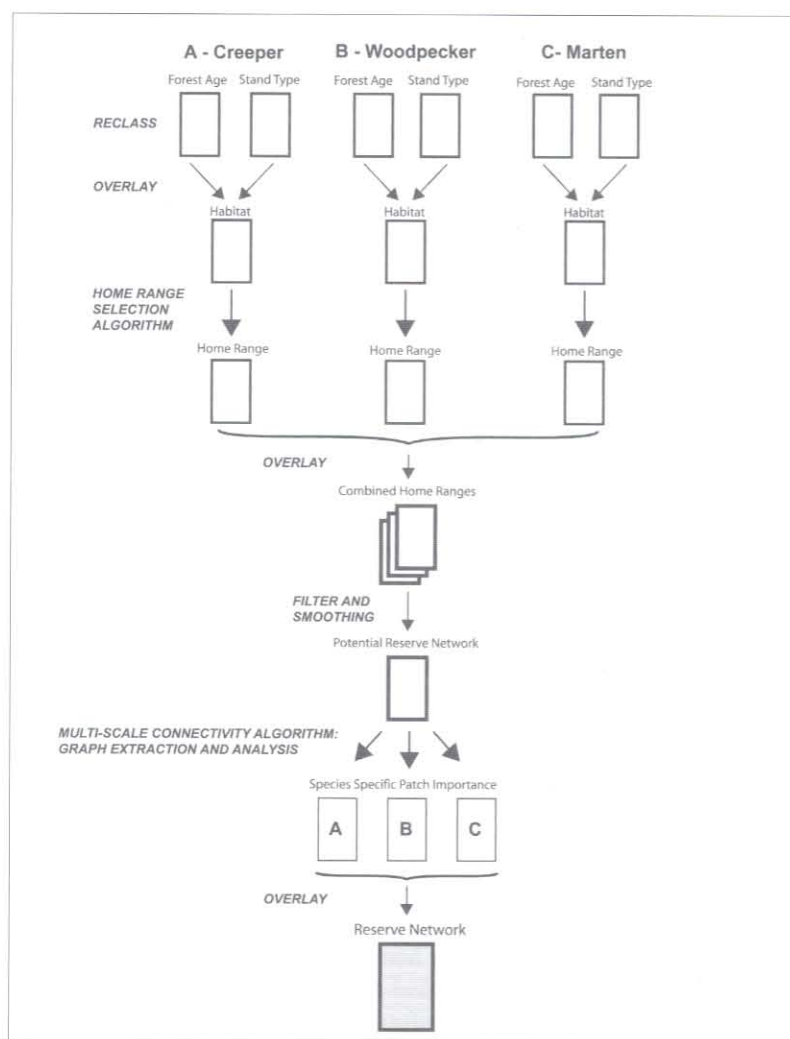


Figure 2. Overall schematic of the spatial reserve selection procedure using three species.

layers and retain cells with the highest values that comprise our conservation target area in a “potential reserve patch” layer. The methods involved in creating the spatial layers in each step of this procedure are as follows:

(1) *Suitable habitat layer*: To identify suitable habitat for each species, maps of forest age and forest stand type were overlaid and reclassified into binary maps of suitable and unsuitable habitat based on species’ preferences listed in Table 1 (see Figure 2). These binary layers comprise the initial stage of analysis from which species’ home ranges will be derived.

(2) *Home range density layer*: To identify areas where the most compact home ranges may be possible, an expanding moving window was applied iteratively across each species’ suitable habitat layer. This method calculates a measure of habitat density surrounding each cell based on a grain size corresponding to a species’ home range

(Figure 3a). Each moving window progressively expands in concentric circles from a focal cell and the number of cells of suitable habitat encountered is counted. The window continues to expand until either: (a) it reaches the maximum home range size specified for the species, or (b) a minimum number of cells of habitat have been captured in the window (Figure 3b). Potential home ranges are identified when the expanding window contains habitat equal to 50% of the maximum home range size. The moving window will stop expanding when this amount of habitat is counted, regardless of window diameter, or when it reaches the maximum home range size. The density of habitat within the home range window is calculated as the percentage of habitat within the home range area and assigned to the focal cell in a new data layer called “home range density.”

(3) *Potential reserve patch layer*: The next step is to use the home range density layer to identify as many non-overlapping home ranges as possible. This step also uses an expanding moving window, but cells that were identified as capable of supporting a home range are processed in decreasing order of habitat density. Home ranges are identified as the moving window expands to include sufficient habitat subject to the species’ spatial constraints outlined above. Expanding home ranges are not permitted to overlap and as such, some of the potential home ranges found in the previous step will not persist due to the priority of higher density home ranges. By processing cells in order of decreasing density, home ranges will be placed preferentially in areas with a higher density of suitable habitat. When a home range is identified, the location of its centroid and its density are recorded. In a post-processing step, the expanding moving window is re-initiated in all home range centroids simultaneously such that as many as possible non-overlapping home ranges can be fit on the landscape. This step reduces pattern artefacts caused by the sequenced identification of potential home ranges. The resultant potential home range layers for each species in our case study are illustrated in Figure 4a.

The next step is to overlay and reclassify the home range maps such that cells with the highest combined habitat density values that comprise our target 12% of the landscape are retained. All other cells are given a value of zero. This layer represents the best sites considering the spatial requirements of each species, existing suitable habitat, as well as the constraints imposed by our target area (Figure 4b). The combined home range layers are illustrated in Figure 4c. The 12% of the landscape selected in this manner is then smoothed using a 3 x 3 maximum filter followed by a 7 x 7 mode filter in

IDRISI. This combined and smoothed map of home ranges made up 23% of the landscape and had persistent habitat patches that were smaller than the home range size of the area-limited species (American marten). Therefore, patches smaller than 500 ha were subsequently removed. The combined, smoothed, and filtered home range density layer represents the potential components of a reserve network (Figure 4d) and was used as input into the spatial connectivity algorithm. The spatial connectivity algorithm assesses the importance of potential reserves sites to maintaining landscape connectivity at multiple spatial scales. The different scales at which connectivity is assessed are representative of the different scales at which the species experience the landscape.

Objective 2: Determine Landscape Connectivity using Spatial Graph Theory

Spatial Graph Theory

Existing methods to quantify connectivity range from data intensive, individual-based simulation movement models [Fahrig and Merriam 1985; Goodwin and Fahrig 2002b] to purely structural landscape pattern metrics based on percolation theory [Gardner *et al.* 1989; Green 1994; With and Crist 1995; With *et al.* 1997; Metzger and Décamps 1997]. Recently, graph theory has also been applied to quantify landscape connectivity wherein a landscape is conceptualized as a mathematical graph consisting of nodes and links among them [Harary 1972]. In this spatial context, graph nodes represent habitat patches and graph links represent the shortest pathways between those patches [Keitt *et al.* 1997; Bunn *et al.* 2000; Ricotta *et al.* 2000; Urban and Keitt 2001; O'Brien *et al.* in press]. Spatial graph theory has been previously applied in an ecological context to examine habitat connectivity for the Mexican Spotted Owl [Keitt *et al.* 1997] and for the Prothonotary Warbler [Bunn *et al.* 2000]. Ricotta *et al.* [2000] use a similar graph theoretic framework to examine the structure of adjacent cover types. Recently, spatial graph theory has also been used to assess the connectivity of habitat for woodland caribou in fragmented forests [O'Brien *et al.* in press].

An accessible general description of graph theory and its applications in ecology can be found in Urban and Keitt [2001]. We do not replicate their review, but build upon their central concepts and

use them to assess landscape connectivity for the purposes of reserve selection. However, we do present some basic definitions regarding graph theory based on Harary's [1972] text and describe the application of graph theory in a spatial context. A graph G is composed of a set of nodes or vertices $V(G)$ and edges $E(G)$ wherein each edge $e_{ij} = v_i v_j$ connects the nodes v_i and v_j . A unique sequence of nodes connected by edges (links) is referred to as a path. A graph is said to be connected if a path exists among each pair of nodes.

The connectivity of a landscape depends not only on the spatial distribution of habitat across a

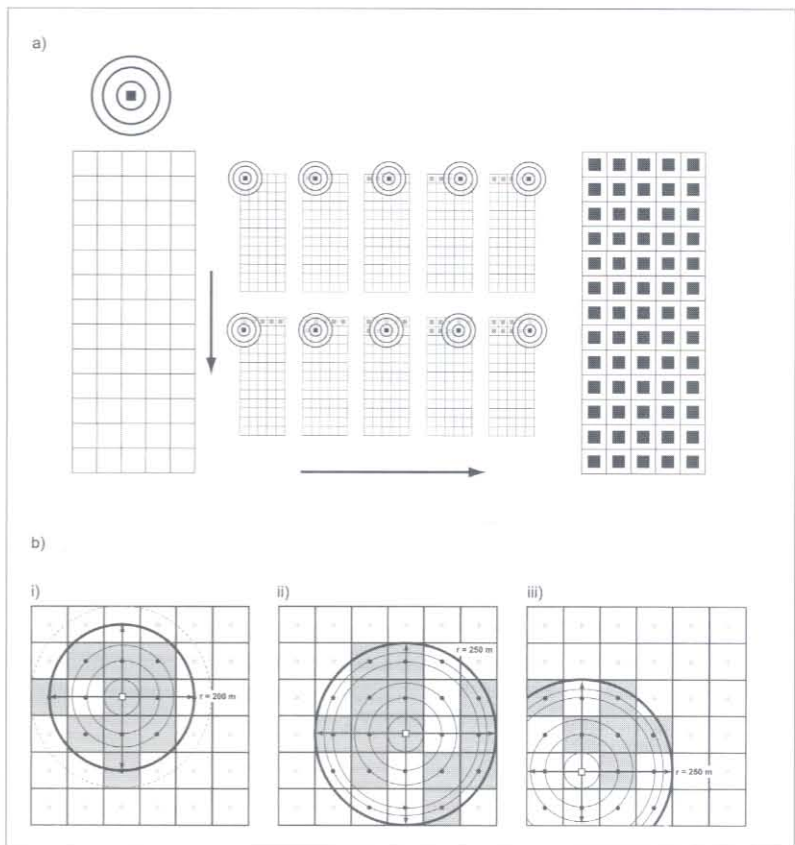


Figure 3. Spatial home range algorithm. (a) The moving window processes all cells in the habitat layer through circular expansion around each focal cell, while counting suitable habitat cells, up to a maximum home range size. This produces a home range density layer wherein the value of each cell represents the density of habitat surrounding it. (b) Illustration of the three possible scenarios of expanding window termination within the home range identification algorithm depicted in (a) using an artificial landscape with grid size of 1 ha/cell and a hypothetical organism with a minimum habitat requirement of 10 ha and a maximum home range size of 20 ha (maximum home range diameter is 250 m). (i) Expansion terminates when minimum amount of habitat required is attained before reaching the maximum home range diameter (dashed circle). (ii) Expansion terminates when the maximum home range diameter is reached where minimum habitat requirement is met. (iii) Expansion terminates when maximum home range diameter is reached; minimum habitat requirements are not met. In the first two cases, habitat density within the home range is calculated and assigned to the focal cell, while in the latter case a value of zero is assigned to the focal cell.

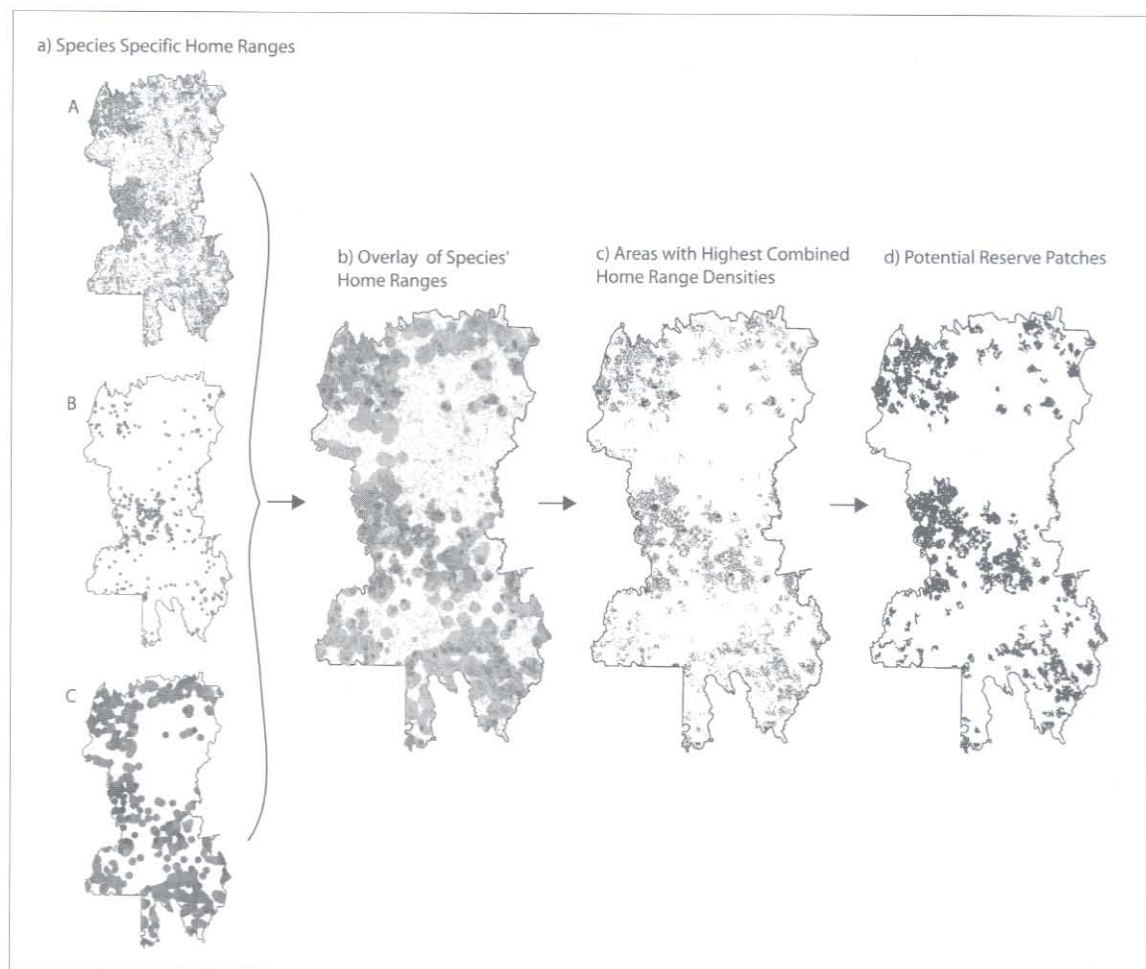


Figure 4. Steps to generate the potential reserve patch layer based on biodiversity representation objective. (a) Spatial home range algorithm results for A – Brown Creeper, B – Pileated Woodpecker, and C – American Marten. Home range densities range from 0% (white) to 100% (dark gray) with a maximum home range density of 100%, 96%, and 93% for the A, B, and C respectively. b) Species specific home range layers are overlaid and summed. c) Cells are ranked by their combined home range values and are retained in decreasing order until 12% of the landscape is captured. d) A binary potential reserve patch layer is created from the high-density home range cells identified in c).

landscape, but also on the scale at which organisms interact with landscape pattern [Merriam 1984; Gardner *et al.* 1989; Noss 1991]. A path may connect a set of nodes for one type of organism, but not for others with lower dispersal capabilities. A graph-theoretic approach to assessing landscape connectivity facilitates a multi-scale analysis of how patches are linked at various distance thresholds (i.e., different species' dispersal capabilities) that can address different organisms' perceptions of connectivity on the same landscape.

Making a correspondence between connections among habitat patches and graph links requires careful consideration of the metric space applied for link distances. Link distances can be represented by Euclidean distances between patches or a measure of

movement cost (i.e. *effective distance* through a cost surface). This is a rich area of research that has much potential for further study, application (e.g., wildlife, forest, transport, ecotourism), and conceptual development (e.g., graph theory, computer science, spatial statistics, spatial decision support system tools). Most past research has simply made use of existing graph theory algorithms.

Although the similarity of patches as graph nodes and inter-patch connections as links is striking, there are a number of key differences between graph theory and spatial graph theory [O'Brien *et al.* in press], including:

- In graph theory, nodes are aspatial (dimensionless) points that may or may not be tied to a reference system. Habitat patches, on the other

hand, are 2-dimensional contiguous areas in a particular geographic location.

- In general, graphs may have an arbitrary number of dimensions, depending on the reference system used for the nodes, whereas patch configurations are usually limited to the 2-dimensional plane.

Application of spatial graph theory presents opportunities to extend and refine standard graph theory methods for the analysis of habitat patch configuration. For example, habitat patches are often identified using raster grids from remotely sensed images or forest inventory maps. Standard graph theory, however, is more directly applicable to point data (vector format). Triangulation algorithms are a basic tool for analyzing sets of points on a plane, and the most commonly used algorithm is the Delaunay triangulation [Okabe *et al.* 1992]. When treating spatial patches as nodes (i.e. areas with various shapes), edge-to-edge links more closely reflect animal perception of the landscape and are more appropriate than centroid-to-centroid links. Therefore, a modified Delaunay triangulation algorithm is used to process spatial patches as nodes using raster data [O'Brien *et al.* in press].

The product of this modified triangulation algorithm is the Minimum Planar Graph (MPG). The MPG is the spatial extension of the Delaunay triangulation and represents the maximum number of non-crossing links among nodes that are of shortest length [O'Brien *et al.* in press]. In a raster context, the distance between a pair of cells (p_1, p_2) is defined by the cost values of the cells traversed between p_1 and p_2 . Here, we use simple Euclidian distance cost such that each cell has a cost value of 1. Between nodes, the cost (distance) of a line ($p_1, p_2, \dots, p_k$) is simply the sum of the costs of each pair of successive points traversed (p_1, p_2), (p_2, p_3), (p_3, p_4), etc. It is important to reiterate here that the links that comprise the MPG connect with cells on the perimeter of a habitat patch, and not with its centroid.

Patch Selection Algorithm

To select a subset of patches for wildlife protection, we assess habitat connectivity using the MPG and a spatial graph algorithm that identifies connected clusters of patches at multiple Euclidean distance thresholds that correspond to our selected species' dispersal capabilities [O'Brien *et al.* in press]. Patches were then prioritized for inclusion in the reserve network based on their contribution to connectivity. The first step in this analysis involves creating the minimum planar graph

(MPG) from the potential reserve patch layer (Figure 6a). The second step is a multi-scale analysis of the MPG that consists of incrementally increasing a distance threshold τ and including only those links with length less than or equal to τ , and considering large connected clusters of patches at each species' dispersal distance threshold (Table 1). For each threshold distance τ , it is assumed that movement is not impeded up to that distance, and all links longer than that distance are removed such that the patches that remain linked are considered connected (a cluster of patches) at this threshold.

The area of habitat connected at a given threshold can be evaluated by calculating a spatial graph theory based metric called expected cluster size (ECS). ECS represents the mean size of a habitat cluster for a randomly selected habitat cell (equivalent to area-weighted mean cluster size; Gustafson [1998]). By computing ECS at each threshold distance, we are able to identify the level of connectivity likely to be experienced by each of the species on the landscape at the threshold distance that corresponds to their dispersal ability (Figure 5). In this example, dispersal ability for each species is conservatively estimated as the diameter of a circle with area equal to maximum home range size.

The selection of patches to include in a spatial reserve network is refined on the basis of an assessment of the importance of each candidate patch in maintaining the connectivity of all patches given the dispersal ability of each species. The status of a

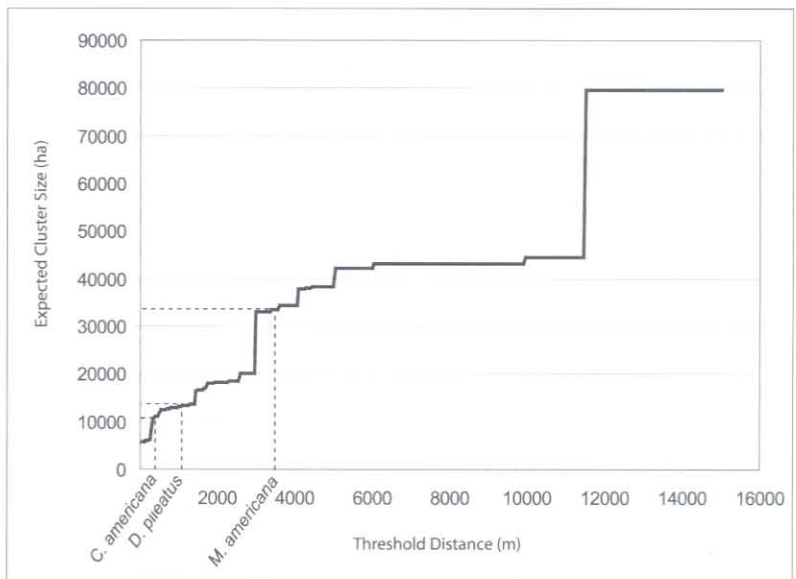


Figure 5. Expected cluster size computed over increasing Euclidean distances from the potential reserve patches. The expected cluster size perceived by each of the species is determined by its dispersal ability (Brown creeper – 226m, Pileated woodpecker – 1572m, American marten 3500m).

patch after such an assessment is interpreted as its biodiversity value and contribution to persistence, which is distinct from its biodiversity content [Sarkar 2002]. At the threshold distances corresponding to the dispersal abilities of the three species, the importance of each patch on the landscape in maintaining connectivity is assessed by computing the deviation in ECS as each patch is successively removed from the network (i.e. the sensitivity of ECS to removal of the patch). Figure 6b shows the results of the patch importance analysis. The overlap among connected habitat patches isolated for each species' particular range of dispersal is used to indicate the priority areas for conservation in this study region. The final priority reserve network layer identified using this multi-scale connectivity analysis is illustrated in Figure 6c.

Conclusions

The spatial method and algorithms presented here integrate diverse types of ecologically relevant information at multiple scales, making it highly useful for informing decisions regarding reserve network selection. The goal of the present study was to illustrate how these methods can assist the selection of reserve networks representative of target levels of biodiversity that will likely persist over time. The objective of biodiversity representation was achieved by using a home range algorithm that is transparent and easy to implement in any GIS, making it practical for delineating species home ranges in other management areas. The spatial graph theory based landscape connectivity

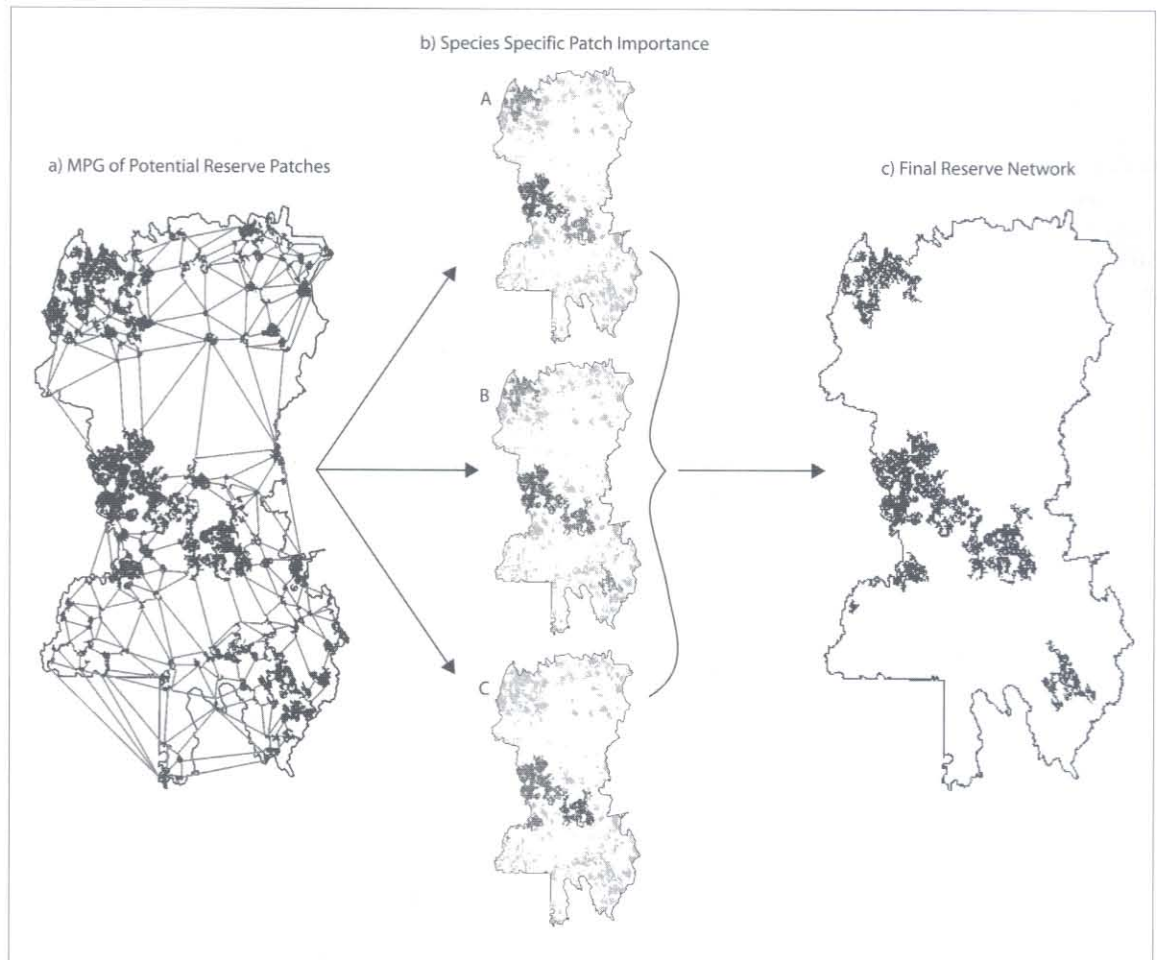


Figure 6. Spatial graph connectivity analysis to identify patches for inclusion in the final reserve network. a) All links in the minimum planar graph (MPG) between potential reserve patches. b) Patch importance analysis performed at multiple scales corresponding to the dispersal abilities of the species: A – Brown Creeper, B – Pileated Woodpecker, C - American Marten. Darker patches signify a higher patch importance value. c) The final reserve network is the overlap of the patches identified as most important in maintaining landscape connectivity for each of the species.

analysis achieved the second objective of planning reserve networks for biodiversity persistence. This combined geomatics (GIS, spatial graph theory) method provides a coherent framework for forest management that can be used to improve the likelihood of biodiversity persistence and reduce the probability of local extinction through the identification of the most important habitat patches for multiple species under a given conservation area budget. We also successfully addressed challenges regarding the quantification of landscape connectivity at different scales in the context of different wildlife species' spatial requirements [Tischendorf and Fahrig 2000].

The tools and species selected for use in the design of reserve networks are highly dependent on specific conservation goals and the characteristics of the study area. There is also a trade-off between capturing requirements for many species and encumbering the analysis process with too much detail. The method as demonstrated is only applicable to situations in which multiple wildlife species share overlapping habitat requirements at different spatial scales. However, many opportunities exist to extend and refine this method beyond the example presented here. The landscape connectivity analytic tools we present are flexible and can be extended to other wildlife species to support many land-use decisions including the location of roads and transmission corridors, the development of forest management plans, and the establishment of reserve areas. Although we calculated the MPG of the potential habitat layer using Euclidean distances, it is also possible to describe network connectivity using least-cost paths based on a movement cost surface that reflects perceived barriers or impediments to wildlife movement [O'Brien *et al.* in press]. Furthermore, the method allows for the integration of other value considerations through preferential weighting of different patches within the network. Although graph-based connectivity tools can assist in the prioritization of key patches for conservation, ultimate reserve membership can be decided upon using any combination of specific compositional and configurational priorities and objectives. Preferential weightings coupled with alternate priorities could also be of use in extending the method to accommodate situations where species' spatial requirements are non-overlapping. Continued development and application of spatially explicit landscape connectivity tools that incorporate multiple species' habitat requirements and the persistence of their populations through time is required to maintain biodiversity and effectively inform sustainable forest management practices.

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